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Lens holder driving device

Abstract

A lens holder driving device includes a first piezoelectric driver configured to move a first lens holder along an optical-axis direction based on the movement of a first piezoelectric element and a second piezoelectric driver configured to move a second lens holder along the optical-axis direction based on the movement of a second piezoelectric element. The first lens holder includes a portion (first left-bearing portion) near a first shaft member and a portion near a second shaft member. The first piezoelectric driver is provided in the first left-bearing portion and contacts the first shaft member. The second lens holder includes a portion near the first shaft member and a portion (second right-bearing portion) near the second shaft member. The second piezoelectric driver is provided in the second right-bearing portion and contacts the second shaft member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority to Japanese Patent Application No. 2021-156049, filed on Sep. 24, 2021, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The disclosures herein generally relate to lens holder driving devices.

2. Description of the Related Art

(3) Conventionally, a device that uses a piezoelectric element to move a lens holder is known (see Patent Document 1).

(4) The aforementioned device is configured such that two lens holders can be moved separately by using two piezoelectric elements.

(5) In the aforementioned device, however, the respective actuators of the two piezoelectric elements are arranged to be in contact with the same drive shaft. Hence, when the two lens holders are moved simultaneously in the configuration, the vibration (movement) of one piezoelectric element may interfere with the vibration (movement) of the other piezoelectric element, and the movements of the two lens holders may be adversely affected.

(6) Therefore, it would be desirable to provide a lens holder driving device that can move two lens holders stably.

RELATED-ART DOCUMENTS

Patent Documents

(7) [Patent Document 1] Japanese Laid-Open Patent Publication No. 2006-098580

SUMMARY OF THE INVENTION

(8) According to one embodiment of the present invention, a lens holder driving device includes a fixed member; a first lens holder configured to hold a first lens body; a second lens holder configured to hold a second lens body arranged to have a same optical axis as the first lens body; a first shaft member and a second shaft member provided in the fixed member and configured to guide the first lens holder and the second lens holder to move in an optical-axis direction with respect to the fixed member; a first piezoelectric driver including a first piezoelectric element and configured to move the first lens holder along the optical-axis direction based on movement of the first piezoelectric element; and a second piezoelectric driver including a second piezoelectric element and configured to move the second lens holder along the optical-axis direction based on movement of the second piezoelectric element. The first lens holder includes a portion near the first shaft member and a portion near the second shaft member. The first piezoelectric driver is provided in the portion near the first shaft member in the first lens holder such that the first piezoelectric driver contacts the first shaft member. The second lens holder includes a portion near the first shaft member and a portion near the second shaft member. The second piezoelectric driver is provided in the portion near the second shaft member in the second lens holder such that the second piezoelectric driver contacts the second shaft member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other objects and further features of the present invention will be apparent from the following

detailed description when read in conjunction with the accompanying drawings, in which:

- (2) FIG. 1A is a perspective view of a lens holder driving device;
- (3) FIG. 1B is an exploded perspective view of the lens holder driving device;
- (4) FIG. 2 is a schematic view of a camera module;
- (5) FIG. 3 is an exploded perspective view of a lower-side member;
- (6) FIG. 4 is an exploded perspective view of movable members which are supported by shaft members;
- (7) FIG. 5 is a view in which a top view of the movable members, which are supported by the shaft-members, is illustrated on the upper side and a view illustrating the right side of the movable members, which are supported by the shaft-members, is illustrated on the lower side;
- (8) FIG. 6A is a perspective view of a piezoelectric driver pressed against the shaft member by a biasing member;
- (9) FIG. 6B is an exploded perspective view of the piezoelectric driver;
- (10) FIG. 7 is a perspective view of the biasing member attached to a lens holder;
- (11) FIG. 8A is a top view of the lens holder driving device;
- (12) FIG. 8B is a cross-sectional view of the lens holder driving device;
- (13) FIG. 9A is an exploded perspective view of a magnetic field generating member before a magnetic member is attached to a fixed magnet;
- (14) FIG. 9B is a perspective view illustrating the assembled state of the magnetic field generating member after the magnetic member is attached to the fixed magnet;
- (15) FIG. 9C is a top view of a magnetic sensor and the magnetic field generating member;
- (16) FIG. 9D is a view illustrating the right side of the magnetic sensor and the magnetic field generating member;
- (17) FIG. 10 is a perspective view of shaft members and movable members;
- (18) FIG. 11A is a front perspective view of the lens holder; and
- (19) FIG. 11B is a rear perspective view of the lens holder.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(20) A lens holder driving device according to the disclosure can move two lens holders more stably.

(21) A lens holder driving device **101** according to an embodiment of the present disclosure will be described with reference to the accompanying drawings. FIG. 1A is a perspective view of the lens holder driving device **101**. FIG. 1B is an exploded perspective view of the lens holder driving device **101**. FIG. 2 is a schematic view of a camera module CM of a camera-equipped portable device on which the lens holder driving device **101** is mounted.

(22) In the example illustrated in FIGS. 1A, 1B, and 2, X1 represents one direction along the X-axis constituting a three-dimensional orthogonal coordinate system and X2 represents the other direction along the X-axis. Further, Y1 represents one direction along the Y-axis constituting the three-dimensional orthogonal coordinate system, and Y2 represents the opposite direction along the Y-axis. In a similar manner, Z1 represents one direction along the Z-axis constituting the three-dimensional orthogonal coordinate system, and Z2 represents the opposite direction along the Z-axis. The X1 side of the lens holder driving device **101** corresponds to the front side (object side) of the lens holder driving device **101**, and the X2 side of the lens holder driving device **101** corresponds to the rear side (image sensor side) of the lens holder driving device **101**. The Y1 side of the lens holder driving device **101** corresponds to the left side of the lens holder driving device **101**, and the Y2 side of the lens holder driving device **101** corresponds to the right side of the lens holder driving device **101**. The Z1 side of the lens holder driving device **101** corresponds to the upper side of the lens holder driving device **101**, and the Z2 side of the lens holder driving device **101** corresponds to the lower side of the lens holder driving device **101**. The above-described configuration of the three-dimensional orthogonal coordinate system also applies throughout the other drawings.

(23) The lens holder driving device **101** is configured to move a lens body LS along an optical axis OA of the lens body LS.

(24) The lens body LS is an example of an optical member and includes one or a plurality of lenses. Typically, the lens body LS is a cylindrical lens barrel including at least one lens, and is configured such that its central axis is along the optical axis OA. As illustrated in FIG. **1B**, the lens body LS includes a first lens body LS1 that forms a zoom lens and a second lens body LS2 that forms a focus lens.

(25) The lens holder driving device **101** is formed such that the lens body LS can be moved along the optical-axis direction by piezoelectric drivers PD (see FIG. **4**) housed in a housing HS. The optical-axis direction includes a direction along the optical axis OA of the lens body LS and a direction parallel to the optical axis OA. More specifically, the lens holder driving device **101** can move the first lens body LS1 along the optical-axis direction and move the second lens body LS2 along the optical-axis direction as indicated by double arrows AR1 and AR2, respectively, in FIGS. **1B** and **2**. That is, the lens holder driving device **101** can move the first lens body LS1 and the second lens body LS2 separately, along their respective optical-axis directions. Note that the optical axis of the first lens body LS1 and the optical axis of the second lens body LS2 are positioned to be collinear (on the optical axis OA).

(26) The housing HS is a part of fixed members FB and includes a lid member **1** and a base member **2**.

(27) As illustrated in FIG. **2**, the lens holder driving device **101** is used in the camera module CM such as a periscope camera module. In the example illustrated in FIG. **2**, the camera module CM mainly includes a mirror MR, a lens body LS, a lens holder driving device **101**, an image sensor IS, and the like. The mirror MR may be a prism. In the example illustrated in FIG. **2**, the mirror MR is configured to provide a flat reflective surface.

(28) Typically, as illustrated in FIG. **2**, the lens holder driving device **101** is arranged at a position farther from an object than the mirror MR, and is configured such that light LT from the object reflected by the mirror MR reaches the image sensor IS through the lens body LS.

(29) The general outline of the lens holder driving device **101** will be described next with reference to FIGS. **1B** and **3**. FIG. **1B** is an exploded perspective view of the lens holder driving device **101**, and illustrates a state in which the lid member **1** is separated from a lower-side member LM. FIG. **3** is an exploded perspective view of the lower-side member LM, and illustrates a state in which movable members MB are separated from the fixed members FB (excluding shaft members **5**). The movable members MB are configured to be moved in the optical-axis direction by the piezoelectric drivers PD while being guided by the shaft members **5**.

(30) As illustrated in FIG. **1B**, the lens holder driving device **101** includes the lid member **1** and the lower-side member LM which form a part of the housing HS (the fixed members FB). The lid member **1** is configured to cover the upper surface of the lower-side member LM. In this embodiment, the lid member **1** is made of a synthetic resin material. However, the lid member **1** may be made of a non-magnetic material such as austenitic stainless steel.

(31) The base member **2** is a member forming a part of the housing HS. In this embodiment, the base member **2** is made of a synthetic resin material similarly to the lid member **1**. However, the base member **2** may be made of a non-magnetic material such as austenitic stainless steel.

(32) More specifically, as illustrated in FIG. **3**, the base member **2** is shaped like a box with a bottom. The box-shaped outer shape of the base member **2** defines a storage portion 2S. The base member **2** includes approximately rectangular-tube-shaped outer walls 2A and an approximately rectangular flat-plate-shaped bottom surface 2B, which is provided so as to be continuous with the respective lower ends (ends on the Z2 side) of the outer walls 2A. As illustrated in FIG. **3**, the outer walls 2A and the bottom surface 2B are formed by separable independent members, and are integrated by using an adhesive or the like. However, the outer walls 2A and the bottom surface 2B may be formed from a single member. The outer walls 2A include a first side plate 2A1, a second

side plate **2A2**, a third side plate **2A3**, and a fourth side plate **2A4**. The first side plate **2A1** and the third side plate **2A3** oppose each other, and the second side plate **2A2** and the fourth side plate **2A4** oppose each other. The second side plate **2A2** and the fourth side plate **2A4** extend perpendicularly to the first side plate **2A1** and the third side plate **2A3**. That is, the first side plate **2A1** and the third side plate **2A3** extend perpendicularly to the second side plate **2A2** and the fourth side plate **2A4**. The first side plate **2A1** includes a through hole **OP1** for receiving the light **LT** from the object reflected by the mirror **MR**. In a similar manner, the third side plate **2A3** includes a through hole **OP2** to allow the light **LT** to reach the image sensor **IS**. The lid member **1** is joined to the base member **2** by using an adhesive, or the like, to form the housing **HS** together with the base member **2**.

(33) Also, as illustrated in FIG. 3, a cutout **2T** through which a terminal portion **10T** of each circuit board **10** may pass is formed in each of the first side plate **2A1** and the third side plate **2A3** of the base member **2**. This configuration allows the terminal portion **10T** of each circuit board **10** to be arranged outside of the housing **HS**.

(34) The lower-side member **LM** includes, as illustrated in FIG. 3, lens holders **3** as the movable members **MB**, the base member **2** and the shaft members **5** as the fixed members **FB**, and position detecting mechanisms **DT**.

(35) Each lens holder **3** is configured to hold the lens body **LS**. In FIG. 3, each lens holder **3** is formed by injection molding a synthetic resin such as a liquid crystal polymer (**LCP**). The lens holders **3** include a first lens holder **3A**, which is configured to hold the first lens body **LS1**, and a second lens holder **3B**, which is configured to hold the second lens body **LS2**. Note that the first lens holder **3A** and the second lens holder **3B** are the same shape and size.

(36) The movable members **MB** will be described in detail with reference to FIGS. 4 and 5. FIGS. 4 and 5 are views illustrating the movable members **MB** supported by the shaft members **5**. More specifically, FIG. 4 is an exploded perspective view in which the movable members **MB** are supported by the shaft members **5**. A top view of the movable members **MB** supported by the shaft members **5** is illustrated on the upper side of FIG. 5, and a view of the right side of the movable members **MB** supported by the shaft members **5** is illustrated on the lower side of FIG. 5. Note that for the sake of clarity in FIGS. 4 and 5, a coarse dotted pattern has been added to the first lens holder **3A** and a fine dotted pattern has been added to the second lens holder **3B**. In addition, the illustration of covers **CV** and the circuit boards **10** have been omitted for the sake of clarity in FIG. 5.

(37) The shaft members **5** include a first shaft member **5A** whose axis (an axis **5AX**) is parallel to the optical axis **OA**, and a second shaft member **5B** whose axis (an axis **5BX**) is parallel to the optical axis **OA**. Hence, the first shaft member **5A** and the second shaft member **5B** extend parallel to each other in the optical-axis direction. In FIG. 3, one end of each shaft member **5** is fitted to a corresponding one of grooves **2G** formed in the third side plate **2A3** of the base member **2**, and the other end of each shaft member **5** is fitted to another corresponding one of the grooves **2G** (not visible in FIG. 3) formed in the first side plate **2A1** of the base member **2**. However, it may be arranged such that one end of each shaft member **5** is inserted through a corresponding one of through holes (not illustrated) formed in the third side plate **2A3** of the base member **2**, and the other end of each shaft member **5** is inserted through another corresponding one of through holes (not illustrated) formed in the first side plate **2A1** of the base member **2**. The shaft members **5** may be fixed to the base member **2** (the third side plate **2A3** and the first side plate **2A1**) with an adhesive. Note that the first shaft member **5A** and the second shaft member **5B** are made of magnetic metal.

(38) Each lens holder **3** includes, as illustrated in FIG. 4, a holder **31** for holding the lens body **LS** and bearing portions **32** for receiving the shaft members **5**. The bearing portions **32** include a left bearing portion **32L** for receiving the first shaft member **5A** and a right bearing portion **32R** for receiving the second shaft member **5B**. The bearing portions **32** function as rotation stoppers for

stopping each lens holder 3 from rotating around the shaft member 5.

(39) More specifically, the first lens holder 3A includes a first holder 31A for holding the first lens body LS1 and first bearing portions 32A for receiving the shaft members 5. The first bearing portions 32A includes the left bearing portion 32L (a first left-bearing portion 32AL) for receiving the first shaft member 5A and the right bearing portion 32R (a first right-bearing portion 32AR) for receiving the second shaft member 5B.

(40) In a similar manner, the second lens holder 3B includes a second holder 31B for holding the second lens body LS2 and second bearing portions 32B for receiving the shaft members 5. The second bearing portions 32B include the left bearing portion 32L (a second left-bearing portion 32BL) for receiving the first shaft member 5A and the right bearing portion 32R (a second right-bearing portion 32BR) for receiving the second shaft member 5B.

(41) More specifically, as illustrated in FIG. 4, the first left-bearing portion 32AL includes first through holes TH1 capable of receiving the first shaft member 5A. In a similar manner, as illustrated in FIG. 4, the second right-bearing portion 32BR includes a second through hole TH2 capable of receiving the second shaft member 5B.

(42) On the other hand, as illustrated in FIG. 4, the first right-bearing portion 32AR includes a first semi-circle cutout CT1, which can receive the second shaft member 5B and opens rightward (Y2 direction). In a similar manner, as illustrated in FIG. 4, the second left-bearing portion 32BL includes a second semi-circle cutout CT2, which can receive the first shaft member 5A and opens leftward (Y1 direction).

(43) A moving magnet 4 is attached to one of the two bearing portions 32 of each lens holder 3. The moving magnets 4, which include a first moving magnet 4A and a second moving magnet 4B, are members provided to restrict the lens holders 3 from rattling around the shaft members 5.

(44) More specifically, as illustrated in FIG. 4, the first moving magnet 4A is attached to the first right-bearing portion 32AR of the first lens holder 3A, and the second moving magnet 4B is attached to the second left-bearing portion 32BL of the second lens holder 3B.

(45) The first moving magnet 4A is configured such that a part of the first right-bearing portion 32AR of the first lens holder 3A can be pressed against the upper surface of the second shaft member 5B from above by using a magnetic attraction force which acts between the first moving magnet 4A and the second shaft member 5B.

(46) The second moving magnet 4B is configured such that a part of the second left-bearing portion 32BL of the second lens holder 3B can be pressed against the upper surface of the first shaft member 5A from above by using a magnetic attraction force which acts between the second moving magnet 4B and the first shaft member 5A.

(47) The cover CV is attached to the other of the two bearing portions 32 of each lens holder 3. That is, the cover CV is attached to the one to which the moving magnet 4 is not attached among the two bearing portions 32 of each lens holder 3.

(48) Each cover CV is a member provided to cover the corresponding piezoelectric driver PD and includes a metal part 14 and a synthetic resin part 15. In FIG. 4, the covers CV include a first cover CV1 which is attached to the first lens holder 3A and a second cover CV2 which is attached to the second lens holder 3B.

(49) More specifically, as illustrated in FIG. 4, the first cover CV1 is attached to the first left-bearing portion 32AL of the first lens holder 3A, and the second cover CV2 is attached to the second right-bearing portion 32BR of the second lens holder 3B. The first cover CV1 includes a first metal part 14A and a first synthetic resin part 15A, and the second cover CV2 includes a second metal part 14B and a second synthetic resin part 15B.

(50) More specifically, the first cover CV1 is insert molded by injecting a synthetic resin into a mold in a state in which the first metal part 14A has been provided as an insert in the mold. Note that the first cover CV1 may be formed by outsert molding. Alternatively, the first cover CV1 may be integrally made of metal or synthetic resin. The second cover CV2 may also be formed in a

manner similar to those described above.

(51) The piezoelectric driver PD will be described with reference to FIGS. 6A and 6B. FIG. 6A is a perspective view of the piezoelectric driver PD which is pressed against the shaft member 5 by a biasing member 13. FIG. 6B is an exploded perspective view of the piezoelectric driver PD.

(52) Each piezoelectric driver PD is configured to move the corresponding lens holder 3 along the optical-axis direction. In this embodiment, each piezoelectric driver PD is an example of a friction drive using the driving system disclosed in U.S. Pat. No. 7,786,648. Each piezoelectric driver PD includes a piezoelectric element 8, a contact member 9, and the circuit board 10.

(53) Each piezoelectric driver PD may be biased inward (in a direction closer to the optical axis OA) by the corresponding biasing member 13 fixed to the lens holder 3 such that the piezoelectric driver is pressed against the shaft member 5. As illustrated in FIGS. 6A and 6B, the biasing member 13 is made of a metal plate and includes portions (inner edge portions BE) corresponding to two nodes ND (see FIG. 6B), which are formed during the bending vibration (the circular motion to be described later) of the piezoelectric element 8. The biasing member 13 is configured such that the inner edge portions BE corresponding to the two nodes ND are brought into contact with the outer surface of the piezoelectric element 8 (the far side of the optical axis OA) via the circuit board 10. The biasing member 13 may be joined to the piezoelectric driver PD by using, for example, an adhesive.

(54) In the example as illustrated in FIG. 4, the piezoelectric drivers PD include a first piezoelectric driver PD1 configured to cause the first lens holder 3A to move along the optical-axis direction and a second piezoelectric driver PD2 configured to cause the second lens holder 3B to move along the optical-axis direction. The first piezoelectric driver PD1 includes a first piezoelectric element 8A, a first contact member 9A, and a first circuit board 10A. The second piezoelectric driver PD2 includes a second piezoelectric element 8B, a second contact member 9B, and a second circuit board 10B.

(55) The biasing members 13 include a first biasing member 13A, which is arranged so as to press the first piezoelectric driver PD1 against the first shaft member 5A, and a second biasing member 13B, which is arranged so as to press the second piezoelectric driver PD2 against the second shaft member 5B.

(56) FIGS. 6A and 6B each illustrate the first piezoelectric driver PD1 that is pressed against the first shaft member 5A by the first biasing member 13A. However, the description to be given hereinafter with reference to FIGS. 6A and 6B is also similarly applicable to the second piezoelectric driver PD2 which is pressed against the second shaft member 5B by the second biasing member 13B. This is because the first piezoelectric driver PD1 and the second piezoelectric driver PD2 have the same configuration.

(57) As illustrated in FIGS. 6A and 6B, the first piezoelectric element 8A extends in the Z-axis direction that is a direction perpendicular to the optical-axis direction (X-axis direction). That is, the first piezoelectric element 8A extends in the Z-axis direction that is a direction perpendicular to the optical axis OA. The first piezoelectric element 8A is configured to implement a bending vibration (circular motion) having the two nodes ND. More specifically, the first piezoelectric element 8A has a two-layer structure including a first layer for implementing a first bending vibration on the X-Z plane and a second layer for implementing a second vibration on the Y-Z plane. When a voltage is applied to a piezoelectric element forming the first layer and a voltage is applied to a piezoelectric element forming the second layer at individual appropriate timings, the first piezoelectric driver PD1 can cause the first piezoelectric element 8A to bend and vibrate (that is, to move in a circular motion) such that a path drawn by the midpoint of the first piezoelectric element 8A is a circular path centered about a rotation axis 8AX in the top view. That is, the first piezoelectric element 8A can move (that is, in a circular motion) such that its midpoint draws a circle. Note that the rotation axis 8AX is parallel to the Z-axis in the example illustrated in FIG. 6B. In addition, applying voltages at appropriate timings allows the first piezoelectric driver PD1 to

switch the direction of movement (direction of rotation) of the midpoint on the circular path between a clockwise direction and a counterclockwise direction when viewed from the Z1 side. That is, the first piezoelectric driver PD1 can switch the direction of movement of the first lens holder 3A along the optical-axis direction. Note that the circle (circular path) drawn by the midpoint of the first piezoelectric element 8A need not be a perfect circle, but suffices to be an approximately circular shape.

(58) In FIG. 6B, a dotted arrow drawn around the first piezoelectric element 8A represents the bending vibration of the first piezoelectric element 8A. That is, the dotted arrow indicates the circular motion in which the first piezoelectric element 8A rotates, while bending, in a clockwise direction about the rotation axis 8AX. In this case, the first piezoelectric driver PD1 that includes the first contact member 9A, which contacts the first shaft member 5A, can move rearward (in the X2 direction). Although not indicated by the dotted arrow, the first piezoelectric element 8A can also rotate, while bending, in a counterclockwise direction about the rotation axis 8AX. In this case, the first piezoelectric driver PD1 that includes the first contact member 9A, which contacts the first shaft member 5A, can move forward (in the X1 direction).

(59) That is, the first lens holder 3A to which the first piezoelectric driver PD1 is attached is moved rearward (in the X2 direction) when the rotational direction of the midpoint of the first piezoelectric element 8A is clockwise in the top view, and is moved forward (in the X1 direction) when the rotational direction of the midpoint of the first piezoelectric element 8A is counterclockwise. Note that as illustrated in FIG. 6B, the midpoint of the first piezoelectric element 8A is a point (a point corresponding to the antinode of the first bending vibration) where the first bending vibration reaches its maximum amplitude, and a point (a point corresponding to the antinode of the second bending vibration) where the second bending vibration reaches its maximum amplitude.

(60) The first contact member 9A is attached to the first piezoelectric element 8A and is configured to be in contact with the first shaft member 5A. In FIG. 6A, the first contact member 9A is adhered to a surface on the inner side of the first piezoelectric element 8A with an adhesive so as to cover the entire surface on the inner side (the Y2 side opposing the optical axis OA) of the first piezoelectric element 8A. The first contact member 9A is made of a metal such as stainless steel, and is formed to have an appropriate thickness that can allow a bending vibration (circular motion) to occur in accordance with the bending vibration (circular motion) of the first piezoelectric element 8A. In FIGS. 6A and 6B, the first contact member 9A is a friction plate made of stainless steel. The first contact member 9A extends in the Z-axis direction which is the same direction as the direction in which the first piezoelectric element 8A extends. The first contact member 9A is formed such that the outer surface (the surface on the Y2 side) of its center portion comes into contact with the first shaft member 5A. More specifically, the first contact member 9A is configured such that the portion (portion corresponding to the antinode of the bending vibration) where the bending vibration (circular motion) reaches its maximum amplitude comes into contact with the first shaft member 5A. In addition, the first contact member 9A has a curved surface (cylindrical shaped surface) on the side (Y2 side) which comes into contact with the first shaft member 5A. The first shaft member 5A is typically made of a metal such as stainless steel. In FIG. 6A, the first shaft member 5A is a stainless-steel cylindrical rod member that extends in the optical-axis direction. Note that as long as the first contact member 9A and the first shaft member 5A are brought into contact with each other, the length of the first contact member 9A in the Z-axis direction may differ from the length of the first piezoelectric element 8A in the Z-axis direction. As illustrated in FIGS. 6A and 6B, the length of the first contact member 9A in the Z-axis direction may be approximately the same as the length of the first piezoelectric element 8A in the Z-axis direction.

(61) The first circuit board 10A is a substrate including a conductive pattern, and is configured to electrically couple the first piezoelectric element 8A to an external power supply. In FIGS. 6A and

6B, the first circuit board **10A** is a flexible printed circuit board, and includes a bonding portion **10F**, a sensor fixing portion **10S**, a piezoelectric element fixing portion **10P**, and a curved portion **10W**.

(62) As illustrated in FIG. **6A**, the outer-side (Y1-side) surface of the bonding portion **10F** of the first circuit board **10A** is fixed to the inner-side (Y2-side) surface of the second side plate **2A2** of the base member **2** with an adhesive. The first circuit board **10A** is configured to deform in accordance with the movement of the first lens holder **3A** in the optical-axis direction, and to apply a voltage to the first piezoelectric element **8A** while moving the position of the curved portion **10W** in accordance with the deformation. More specifically, the curved portion **10W** moves rearward as the first lens holder **3A** moves rearward (X2 direction), and moves forward as the first lens holder **3A** moves forward (X1 direction). Furthermore, the first circuit board **10A** is configured such that the inner-side (Y2-side) surface of the piezoelectric element fixing portion **10P** is bonded to the first piezoelectric element **8A** by an anisotropic conductive adhesive, an anisotropic conductive adhesive film, or the like. The first circuit board **10A** is folded back on the front side of the sensor fixing portion **10S** such that the sensor fixing portion **10S** is positioned outside of (Y1 side) of the first biasing member **13A**. The first circuit board **10A** is configured such that a first magnetic sensor **6A** (a magnetic sensor **6**) (See FIG. **3**) is mounted on the outer-side (Y1-side) surface of the sensor fixing portion **10S**. Note that the inner-side (Y2-side) surface of the sensor fixing portion **10S** is fixed to the outer-side (Y1-side) surface of the first cover **CV1** with an adhesive (See FIGS. **8A** and **8B**).

(63) As illustrated in FIGS. **6A** and **6B**, the biasing member **13** is formed by a flat spring member. More specifically, as illustrated in FIG. **6B**, the first biasing member **13A** includes fixing portions **13F** which are fixed to the first left-bearing portion **32AL** of the first lens holder **3A**, support portions **13S** which support the first piezoelectric driver **PD1**, and elastic deforming portions **13E** which are provided between the fixing portions **13F** and the support portions **13S** and are capable of being elastically deformed. The first biasing member **13A** is fixed to the first left-bearing portion **32AL** via the fixing portions **13F** such that the support portions **13S** and the elastic deforming portions **13E** do not come into contact with the first left-bearing portion **32AL**. Although the following description referring to FIG. **6B** is given for the first biasing member **13A**, the following description is also applicable to the second biasing member **13B**. This is because the second biasing member **13B** is the same shape and size as the first biasing member **13A**.

(64) More specifically, the fixing portions **13F** include a front fixing portion **13FF** and a rear fixing portion **13FB**, and the support portions **13S** include an upper support portion **13SU** and a lower support portion **13SD**. The elastic deforming portions **13E** includes an upper elastic deforming portion **13EU** and a lower elastic deforming portion **13ED** provided between the rear fixing portion **13FB** and the front fixing portion **13FF**. The front fixing portion **13FF** and the rear fixing portion **13FB** are the same shape and size. The upper support portion **13SU** and the lower support portion **13SD** are the same shape and size. The upper elastic deforming portion **13EU** and the lower elastic deforming portion **13ED** are the same shape and size. That is, the first biasing member **13A** is arranged to be plane-symmetrical to a plane of symmetry (a plane dividing the first biasing member **13A** into front and rear halves) that is parallel to the Y-Z plane. In addition, the biasing member **13A** is arranged to be plane-symmetrical to another plane of symmetry (a plane dividing the first biasing member **13A** into upper and lower halves) that is parallel to the X-Y plane. The second biasing member **13B** is also arranged to have plane symmetries in a similar manner.

(65) Each support portion **13S** is bent in an L-shape from the corresponding elastic deforming portion **13E** and protrudes to the side (Y2 side) on which the lens holder **3** is positioned. A recessed portion **RS** is formed at the distal end of each support portion **13S**. The recessed portion **RS** is a recessed portion with an opening on the side (Y2 side) on which the first lens holder **3A** is positioned. More specifically, the recessed portions **RS** are formed to be the same shape and size at the respective distal ends of the upper support portion **13SU** and the lower support portion **13SD**.

As illustrated in FIG. 6A, parts of the first piezoelectric driver PD1 are arranged inside the recessed portions RS and are fixed to the support portions 13S with an adhesive while in contact with the respective inner edges BE of the recessed portions RS.

(66) More specifically, as illustrated in FIG. 6B, each recessed portion RS includes a front edge and a rear edge which oppose each other and have the inner edge BE interposed therebetween. The first piezoelectric driver PD1 is arranged between the front edge and the rear edge as illustrated in FIG. 6A.

(67) The respective positions where the inner edges BE of the recessed portions RS come into contact with the first piezoelectric driver PD1 correspond to the respective positions of the nodes ND of the first piezoelectric element 8A which implement the bending vibration (circular motion). The positions of the nodes ND include the position of a first node ND1 and the position of a second node ND2. For the sake of clarity, a cross-hatched pattern has been added to the positions of the nodes ND in FIG. 6B.

(68) Each position (the position of each node ND) where the first piezoelectric driver PD1 comes into contact with the inner edge portion BE of the recessed portion RS corresponds to a position at a predetermined distance from the end portion of the first piezoelectric driving portion PD1. The predetermined distance is, for example, approximately $\frac{1}{4}$ of the total length of the piezoelectric driver PD.

(69) The first piezoelectric driver PD1 is fixed to the support portions 13S with an adhesive. More specifically, the first piezoelectric driver PD1 (the first circuit board 10A) and each support portion 13S are fixed to each other at the inner edge BE of the corresponding recessed portion RS with an adhesive. The first circuit board 10A of the first piezoelectric driver PD1 and the first biasing member 13A are fixed to each other at a contact portion AF of the front fixing portion 13FF and a contact portion AB of the rear fixing portion 13FB with an adhesive. In FIGS. 6A and 6B, the adhesive may be a UV-cured adhesive. However, the adhesive may be another type of adhesive such as a moisture-cured adhesive or a heat-cure adhesive.

(70) As illustrated in FIG. 6B, each elastic deforming portion 13E includes a portion extending forward (in the X1 direction) from the corresponding support portion 13S and a portion extending rearward (in the X2 direction) from the corresponding support portion 13S. More specifically, the upper elastic deforming portion 13EU includes a portion extending forward from the upper support portion 13SU and a portion extending rearward from the upper support portion 13SU, and the lower elastic deforming portion 13ED includes a portion extending forward from the lower support portion 13SD and a portion extending rearward from the lower support portion 13SD. Furthermore, the direction in which each elastic deforming portion 13E extends is along the optical-axis direction.

(71) The fixing portions 13F are provided at both ends of elastic deforming portions 13E. As illustrated in FIG. 7, the fixing portions 13F are attached so as to enclose the upper ends and the lower ends of the bearing portion 32 (the second right-bearing portion 32BR) of the lens holder 3 (the second lens holder 3B).

(72) FIG. 7 is a perspective view of the biasing member 13 attached to the lens holder 3. More specifically, FIG. 7 is a perspective view of the second biasing member 13B attached to the second lens holder 3B. The description to be given hereinafter with reference to FIG. 7 is similarly applicable to the first biasing member 13A which is attached to the first lens holder 3A. This is because the first biasing member 13A and the second biasing member 13B have the same configuration. Note that for the sake of clarity, a cross-hatched pattern has been added to the biasing member 13 (the second biasing member 13B) in FIG. 7.

(73) In the example illustrated in FIG. 7, the fixing portions 13F of the second biasing member 13B include the rear fixing portion 13FB and the front fixing portion 13FF. The rear fixing portion 13FB and the front fixing portion 13FF each include an upper bended portion UB and a lower bended portion DB (see also FIG. 6A). The second biasing member 13B is formed such that the

right end portions of the second bearing portion **32BR** of the second lens holder **3B** can be enclosed by the pair of upper bended portions **UB** and the pair of lower bended portions **DB**. Note that the second biasing member **13B** may be fixed to the second right-bearing portion **32BR** with an adhesive or may be reinforced with an adhesive.

(74) The second biasing member **13B** which is fixed to the right end (the end on the **Y2** side) of the second right-bearing portion **32BR** of the second lens holder **3B** is covered by the second cover **CV2**. The second piezoelectric driver **PD2** which is fixed to the inner side (the **Y1** side) of the second biasing member **13B** is arranged in a second space **SP2** between a second front wall portion **FW2** and a second rear wall portion **BW2** of the second right-bearing portion **32BR**. The second space **SP2** is sealed by the second cover **CV2**.

(75) In FIG. 7, the second cover **CV2** is fixed to the second right-bearing portion **32BR** by fitting rectangular protrusions **PT**, which are formed on the upper end portions of the second right-bearing portion **32BR**, into cutouts **CU**, which are formed on an upper plate portion of the second metal part **14B**. Furthermore, round protrusions (not visible in FIG. 7), which are formed on the lower end portions of the second right-bearing portion **32BR**, are fitted into round holes **BH** which are formed on the lower plate portion of the second metal part **14B**. This configuration is similarly applicable to the first cover **CV1**.

(76) In this configuration, the piezoelectric driver **PD** can be housed in a space sealed by the lens holder **3** and the cover **CV**. Hence, this configuration can prevent debris that is generated due the contact between the shaft member **5** and the contact member **9** from escaping outside of the sealed space. Consequently, the debris can be prevented from reaching the image sensor **IS**. Note that the second cover **CV2** suffices to seal the second space **SP2** such that debris is prevented from escaping outside of the second space **SP2**, and need only seal the space to an extent that allows air to flow in and out from the space.

(77) The movable members **MB** and the position detecting mechanisms **DT** will be described next with reference to FIGS. 3, 8A, and 8B. FIG. 8A is a top view of the lens holder driving device **101**. FIG. 8B is a cross-sectional view of the lens holder driving device **101**. More specifically, FIG. 8B is a view in which the cross-section of the lens holder driving device **101**, which is taken along a plane parallel to the **Y-Z** plane including a cutting plane line (broken lines **VIIIB-VIIIB**) illustrated in FIG. 8A, is seen from the **X1** side. Note that for the sake of clarity in FIGS. 8A and 8B, the lid member **1** and the base member **2** have not been illustrated, a coarse dot pattern has been added to the first lens holder **3A**, and a fine dot pattern has been added to the second lens holder **3B**.

(78) As illustrated in FIG. 8A, the movable members **MB** include a first movable member **MB1** and a second movable member **MB2**. The first movable member **MB1** includes the first lens holder **3A**, the first moving magnet **4A**, the first magnetic sensor **6A**, the first cover **CV1**, and the first lens body **LS1**. The second movable member **MB2** includes the second lens holder **3B**, the second moving magnet **4B**, a second magnetic sensor **6B**, the second cover **CV2**, and the second lens body **LS2**.

(79) As illustrated in FIG. 8B, the first moving magnet **4A** is provided in a position away from a virtual plane **VP**. More specifically, the first moving magnet **4A** is provided in a position not on the virtual plane **VP** (that is, a position higher than the position of the virtual plane **VP**). The virtual plane **VP** is a virtual plane which includes the axis (the axis **5AX**) of the first shaft member **5A** and the axis (the axis **5BX**) of the second shaft member **5B** which are parallel to each other. In the example illustrated in FIG. 8B, each lens holder **3** is arranged such that the optical axis **OA** of the lens body **LS** is positioned on the virtual plane **VP**. Also, in the example illustrated in FIG. 8B, each movable member **MB** is arranged such that the distance between the optical axis **OA** and the axis (the axis **5AX**) of the first shaft member **5A** is equal to the distance between the optical axis **OA** and the axis (the axis **5BX**) of the second shaft member **5B**. In addition, in the example illustrated in FIG. 8B, the first movable member **MB1**, which includes the first lens holder **3A**, the first moving magnet **4A**, the first cover **CV1**, the first lens body **LS1**, and the first piezoelectric driver

PD1, is arranged such that its center of mass is positioned on the virtual plane VP. Arranging the first movable member MB1 in such a manner can limit the torque generated about the center of mass when the first movable member MB1 moves along the shaft members 5. This configuration is similarly applicable to the second movable member MB2. More specifically, as illustrated in FIG. 3, the base member 2 as the fixed member FB includes the bottom surface 2B. As illustrated in FIG. 4, the upper part of each lens holder 3 is opened such that the lens body LS can be accommodated (received). Also, as illustrated in FIG. 8B, the moving magnet 4 (the first moving magnet 4A) is provided in an offset position which is closer to the upper side (Z1 side) than the virtual plane VP. Note that the moving magnet 4 may be provided in an offset position which is closer to the lower side (the Z2 side) than the virtual plane VP.

(80) As illustrated in FIG. 8B, each position detecting mechanism DT is a mechanism for detecting the position of the lens holder 3, and includes the magnetic sensor 6 and a magnetic field generating member MG. The magnetic field generating member MG includes a fixed magnet 11 and magnetic members 12.

(81) Each magnetic sensor 6 is configured to detect the magnetic field generated by the magnetic field generating member MG. In FIG. 8B, the magnetic sensor 6 is formed by a giant magnetoresistive (GMR) element. The magnetic sensor 6 is configured to detect the position of the lens holder 3, to which it is attached, by measuring the voltage value which changes in accordance with the magnitude of the magnetic field received from the magnetic field generating member MG. The magnetic sensor 6 is configured to output a voltage such that the voltage value increases as the N-pole portion becomes closer and the voltage value decreases as the S-pole portion becomes closer. However, the magnetic sensor 6 may be configured to output a lower voltage value as the N-pole portion becomes closer, and a higher voltage value as the S-pole portion becomes closer. In the magnetic sensor 6 may be configured to detect the position of the lens holder 3 by using another magnetoresistive element, such as a semiconductor magnetoresistive (SMR) element, an anisotropic magnetoresistive (AMR) element, or a tunnel magnetoresistive (TMR) element. Alternatively, the magnetic sensor 6 may be configured to detect the position of the lens holder 3 by using a hall sensor or the like.

(82) As illustrated in FIG. 8B, the position detecting mechanisms DT include a first position detection mechanism DT1 for detecting the position of the first lens holder 3A and a second position detection mechanism DT2 for detecting the position of the second lens holder 3B. The first position detection mechanism DT1 includes the first magnetic sensor 6A and a first magnetic field generating member MG1, and the second position detection mechanism DT2 includes the second magnetic sensor 6B and a second magnetic field generating member MG2.

(83) As illustrated in FIG. 8A, a width WD1 is the length (width) of the first magnetic field generating member MG1 in the optical-axis direction, and a width WD2 is the length (width) of the second magnetic field generating member MG2 in the optical-axis direction. The first magnetic field generating member MG1 overlaps the second magnetic field generating member MG2 over a distance DS in the optical-axis direction.

(84) In addition, as illustrated in FIG. 8B, the first magnetic field generating member MG1 includes a first fixed magnet 11A and first magnetic members 12A, and the second magnetic field generating member MG2 includes a second fixed magnet 11B and second magnetic members 12B. The first magnetic members 12A include an upper first magnetic member 12AU and a lower first magnetic member 12AD, and the second magnetic members 12B include an upper second magnetic member 12BU and a lower second magnetic member 12BD.

(85) The magnetic field generating member MG will be described in detail with reference to FIGS. 9A to 9D. FIG. 9A is an exploded perspective view of the magnetic field generating member MG before the magnetic members 12 are attached to the fixed magnet 11. FIG. 9B is a perspective view illustrating the assembled state of the magnetic field generating member MG after the magnetic members 12 were attached to the fixed magnet 11. FIGS. 9C and 9D each illustrate the positional

relationship between the magnetic sensor **6** and the magnetic field generating member MG which changes in accordance with the movement of the lens holder **3** in the optical-axis direction. More specifically, FIG. **9C** is a top view of the magnetic sensor **6** and the magnetic field generating member MG, and FIG. **9D** is a view illustrating the right side of the magnetic sensor **6** and the magnetic field generating member MG. Note that FIGS. **9A** to **9D** are related to the first magnetic field generating member MG1 which is attached to the inner surface of the second side plate **2A2** of the base member **2**. In FIGS. **9A** to **9D**, assume that the first magnetic field generating member MG1 is fixed to the inner surface of the second side plate **2A2** of the base member **2** with an adhesive. Note that the following description to be given with reference to FIGS. **9A** to **9D** is also similarly applicable to the second magnetic field generating member MG2 which is attached to the inner surface of the fourth side plate **2A4** of the base member **2**.

(86) As illustrated in FIG. **9A**, the first magnetic field generating member MG1 includes the first fixed magnet **11A**, the upper first magnetic member **12AU**, and the lower first magnetic member **12AD**.

(87) The first fixed magnet **11A** is a permanent magnet that has undergone dipole magnetization in the Z-axis direction which is a direction perpendicular to the optical-axis direction. In FIG. **9A**, the first fixed magnet **11A** has been magnetized such that one end (the upper end portion) in the direction (Z-axis direction) perpendicular to the optical-axis direction is the N pole and the other end (the lower end portion) is the S pole. For the sake of clarity, a fine cross-hatched pattern has been added to the N-pole portion of the first fixed magnet **11A** and a coarse cross-hatched pattern has been added to the S-pole portion of the first fixed magnet **11A** in FIG. **9A**.

(88) In FIG. **9B**, both the upper first magnetic member **12AU** and the lower first magnetic member **12AD** have been coupled to the first fixed magnet **11A**. In FIG. **9B**, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** are fixed to the first fixed magnet **11A** with an adhesive.

(89) More specifically, the upper first magnetic member **12AU** is magnetized to the N pole by being coupled to the N-pole portion which is the one end (the upper end portion) of the first fixed magnet **11A**, and the lower first magnetic member **12AD** is magnetized to the S pole by being coupled to the S-pole portion which is the other end (the lower end portion) of the first fixed magnet **11A**. Hence, in FIG. **9B**, a fine cross-hatched pattern has been added to the upper first magnetic member **12AU** in a manner similar to the N-pole portion of the first fixed magnet **11A**, and a coarse cross-hatched pattern has been added to the lower first magnetic member **12AD** in a manner similar to the S-pole portion of the first fixed magnet **11A**. Fine and coarse cross-hatched patterns have been illustrated in FIGS. **9C** and **9D** in a similar manner. However, in the state illustrated in FIG. **9A**, the fine cross-hatched pattern and coarse cross-hatched pattern have not been added to the upper first magnetic member **12AU** and the lower first magnetic member **12AD**, respectively, because neither of the upper first magnetic member **12AU** and the lower first magnetic member **12AD** is magnetized by being coupled to the first fixed magnet **11A**.

(90) As illustrated in FIG. **9A**, each first magnetic member **12A** includes a rod-shaped portion BP and L-shaped protrusions PR which protrude from the rod-shaped portion BP in the Y-axis direction and subsequently protrude in the Z-axis direction. More specifically, the upper first magnetic member **12AU** includes an upper rod-shaped portion BPU and three protrusions PR (a first upper protrusion PR1U to a third upper protrusion PR3U). Each of the first upper protrusion PR1U to the third upper protrusion PR3U protrudes rightward (the Y2 direction) from the upper rod-shaped portion BPU and subsequently protrudes toward the side of the first fixed magnet **11A** (the Z2 direction). The lower first magnetic member **12AD** includes a lower rod-shaped portion BPD and three protrusions PR (a first lower protrusion PR1D to a third lower protrusion PR3D). Each of the first lower protrusion PR1D to the third lower protrusion PR3D protrudes rightward (the Y2 direction) from the lower rod-shaped portion BPD and subsequently protrudes toward the side of the first fixed magnet **11A** (the Z1 direction).

(91) Furthermore, in FIG. 9A, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** are formed such that the first upper protrusion **PR1U** to the third upper protrusion **PR3U** and the first lower protrusion **PR1D** to the third lower protrusion **PR3D** are arranged alternately in the extending direction (the X-axis direction) of the first magnetic field generating member **MG1**.

(92) More specifically, as illustrated in FIG. 9B, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** are formed such that the first upper protrusion **PR1U** is arranged on the frontmost side (X1 side) of the extending direction (X-axis direction) of the first magnetic field generating member **MG1**, the first lower protrusion **PR1D** is arranged on the rear side (X2 side) of the first upper protrusion **PR1U**, the second upper protrusion **PR2U** is arranged further on the rear side (X2 side) of the first lower protrusion **PR1D**, the second lower protrusion **PR2D** is arranged further on the rear side (X2 side) of the second upper protrusion **PR2U**, the third upper protrusion **PR3U** is arranged further on the rear side (X2 side) of the second lower protrusion **PR2D**, and the third lower protrusion **PR3D** is arranged further on the rear side (X2 side) of the third upper protrusion **PR3U**.

(93) In addition, as illustrated in FIG. 9A, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** are formed such that a width **M1** of the first upper protrusion **PR1U**, a width **M2** of a second upper protrusion **PR2U**, a width **M3** of a third upper protrusion **PR3U**, a width **M4** of the first lower protrusion **PR1D**, a width **M5** of a second lower protrusion **PR2D**, and a width **M6** of the third lower protrusion **PR3D** in the extending direction (X-axis direction) of the first magnetic field generating member **MG1** are the same.

(94) In addition, as illustrated in FIG. 9A, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** are formed such that a space **L1** between the first upper protrusion **PR1U** and the second upper protrusion **PR2U**, a space **L2** between the second upper protrusion **PR2U** and the third upper protrusion **PR3U**, a space **L3** between the first lower protrusion **PR1D** and the second lower protrusion **PR2D**, and a space **L4** between the second lower protrusion **PR2D** and the third lower protrusion **PR3D** in the extending direction (X-axis direction) of the first magnetic field generating member **MG1** are the same length.

(95) Further, as illustrated in FIG. 9B, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** are formed such that a space **N1** between the first upper protrusion **PR1U** and the first lower protrusion **PR1D**, a space **N2** between the first lower protrusion **PR1D** and the second upper protrusion **PR2U**, a space **N3** between the second upper protrusion **PR2U** and the second lower protrusion **PR2D**, a space **N4** between the second lower protrusion **PR2D** and the third upper protrusion **PR3U**, and a space **N5** between the third upper protrusion **PR3U** and the third lower protrusion **PR3D** in the extending direction (the X-axis direction) of the first magnetic field generating member **MG1** are the same length.

(96) In addition, the first upper protrusion **PR1U** to the third upper protrusion **PR3U** of the upper first magnetic member **12AU** and the first lower protrusion **PR1D** to the third lower protrusion **PR3D** of the lower first magnetic member **12AD** include protruding portions in the Z-axis direction (the side of the first fixed magnet **11A**). As illustrated in FIG. 9C, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** are formed such that the respective thicknesses **TK** of the protruding portions in the Z-axis direction are the same.

(97) Furthermore, as illustrated in FIG. 9C, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** are formed such that a uniform space **GP** is provided between the magnetic sensor **6** (the first magnetic sensor **6A**), which is fixed to the lens holder **3** (not illustrated), and the right-side (Y2-side) end of each of the first upper protrusion **PR1U** to the third upper protrusion **PR3U** and the first lower protrusion **PR1D** to the third lower protrusion **PR3D**.

(98) In this manner, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** are formed as identical components that are the same shape and size. However, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** may be formed as

different components that are different shapes or sizes. For example, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** may be formed to have varying widths **M1** to **M6**. Further, the spaces **L1** to **L4** may be formed to have varying lengths in a similar manner, and the spaces **N1** to **N5** may also be formed to have varying lengths in a similar manner. Furthermore, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** may be formed to have varying thicknesses **TK** in respective portions, which protrude in the Z-axis direction (the side of the first fixed magnet **11A**), of the first upper protrusion **PR1U** to the third upper protrusion **PR3U** and the first lower protrusion **PR1D** to the third lower protrusion **PR3D**. The space **GP** may also vary in a similar manner.

(99) In the embodiment, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** are formed such that the respective distal end portions, which protrude in the Z-axis direction (the side of the first fixed magnet **11A**), of the first upper protrusion **PR1U** to third upper protrusion **PR3U** and the first lower protrusion **PR1D** to third lower protrusion **PR3D** reach the boundary of the N-pole portion and the S-pole portion of the first fixed magnet **11A** in a view seen from the right side as illustrated in FIG. **9D**. However, the upper first magnetic member **12AU** and the lower first magnetic member **12AD** may be formed to have varying amounts of protrusion with respect to the respective portions, which protrude in the Z-axis direction (the side of the first fixed magnet **11A**), of the first upper protrusion **PR1U** to third upper protrusion **PR3U** and the first lower protrusion **PR1D** to third lower protrusion **PR3D**.

(100) The positional relationship between the first magnetic field generating member **MG1** and the first magnetic sensor **6A** when the first lens holder **3A** is positioned at a reference position indicated in FIG. **1B** is illustrated in the center portion of each of FIGS. **9C** and **9D**. The positional relationship between the first magnetic field generating member **MG1** and the first magnetic sensor **6A** when the first lens holder **3A** has moved forward (in the **X1** direction) from the reference position is illustrated in the upper portion of each of FIGS. **9C** and **9D**. The positional relationship between the first magnetic field generating member **MG1** and the first magnetic sensor **6A** when the first lens holder **3A** moves rearward (in the **X2** direction) from the reference position is illustrated in the lower portion of each of FIGS. **9C** and **9D**.

(101) To detect the magnetic field generated by each of the first upper protrusion **PR1U** to the third upper protrusion **PR3U** and the first lower protrusion **PR1D** to the third lower protrusion **PR3D**, the first magnetic sensor **6A** is provided on the sensor fixing portion **10S** of the first circuit board **10A** fixed to the first lens holder **3A**.

(102) As illustrated in the center portion of each of FIGS. **9C** and **9D**, when the first lens holder **3A** is positioned at the reference position, the first magnetic sensor **6A** partially opposes the second upper protrusion **PR2U** of the upper first magnetic member **12AU** as the N-pole portion.

(103) As illustrated in the upper portion of each of FIGS. **9C** and **9D**, when the first lens holder **3A** moves forward (in the **X1** direction) from the reference position, the first magnetic sensor **6A** moves away from the second upper protrusion **PR2U** of the upper first magnetic member **12AU** as the N-pole portion to partially face the first lower protrusion **PR1D** of the lower first magnetic member **12AD** as the S-pole portion.

(104) Also, as illustrated in the lower portion of each of FIGS. **9C** and **9D**, when the first lens holder **3A** moves rearward (in the **X2** direction) from the reference position, the first magnetic sensor **6A** moves away from the second upper protrusion **PR2U** of the upper first magnetic member **12AU** as the N-pole portion to partially face the second lower protrusion **PR2D** of the lower first magnetic member **12AD** as the S-pole portion.

(105) By using the position detecting mechanisms **DT** as described above, a control device (not illustrated), which is coupled to the first magnetic sensor **6A** via a conductive pattern formed on the first circuit board **10A**, can detect the position of the first lens holder **3A** based on the value of the voltage output from the first magnetic sensor **6A**. Note that although the control device is provided outside the housing **HS** in the embodiment, the control device may be provided inside the housing

HS.

(106) In the embodiment, each magnetic field generating member MG is formed by a single dipole permanent magnet. However, each magnetic field generating member MG may be formed by a combination of a plurality of dipole permanent magnets, a single multipole permanent magnet, or a combination of a plurality of multipole permanent magnets.

(107) The movement of the movable members MB will be described with reference to FIG. 10. FIG. 10 is a perspective view of the shaft members 5 and the movable members MB. More specifically, the positional relationship of the main components (the lens holders 3, the shaft members 5, and the lens bodies LS) when the first movable member MB1 and the second movable member MB2 are at their respective reference positions is illustrated in the upper part of FIG. 10. The positional relationship of the main components when the first movable member MB1 has moved to the frontmost side (the X1 side) and the second movable member MB2 has moved to the frontmost side (the X1 side) is illustrated in the middle part of FIG. 10. The positional relationship of the main components when the second movable member MB2 moves to the rearmost side (the X2 side) and the first movable member MB1 moves to the rearmost side (the X2 side) is illustrated in the lower part of FIG. 10.

(108) As illustrated in FIG. 10, the lens holder driving device 101 can move the first movable member MB1 and the second movable member MB2 separately along the optical-axis direction.

(109) The lens holder 3 will be described in detail next with reference to FIGS. 11A and 11B. FIG. 11A is a front perspective view of the lens holder 3, and FIG. 11B is rear perspective view of the lens holder 3. Although the following description to be given with reference to FIGS. 11A and 11B is related to the first lens holder 3A, the following description can also be similarly applied to the second lens holder 3B.

(110) As described above, the first lens holder 3A includes the first holder 31A, the first left-bearing portion 32AL, and the first right-bearing portion 32AR.

(111) The first holder 31A includes a first through hole RH1 through which the optical axis OA passes. The first through hole RH1 include a first front through hole RHF1 and a first rear through hole RHB1. In FIGS. 11A and 11B, the first front through hole RHF1 is formed by a pair of U-shaped grooves, that is, a lower front U-shaped groove DCF which opens upward in the Z1 direction (predetermined direction) and an upper front U-shaped groove UCF which opens downward in the Z2 direction (direction opposite to the predetermined direction). More specifically, the lower front U-shaped groove DCF and the upper front U-shaped groove UCF are arranged to be adjacent to each other in the front to rear direction (the X-axis direction) and to oppose each other in the vertical direction. In a similar manner, the first rear through hole RHB1 is formed by a pair of U-shaped grooves, that is, a lower rear U-shaped groove DCB which opens upward in the Z1 direction (predetermined direction) and an upper rear U-shaped groove UCB which opens downward in the Z2 direction (direction opposite to the predetermined direction). More specifically, the lower rear U-shaped groove DCB and the upper rear U-shaped groove UCB are arranged to be adjacent to each other in the front to rear direction (the X-axis direction) and to oppose each other in the vertical direction.

(112) In addition, the first left-bearing portion 32AL includes first through holes TH1 through which the axis 5AX of the first shaft member 5A passes. The first through holes TH1 include a first front through hole THF1 and a first rear through hole THB1. In FIG. 11B, the first front through hole THF1 is formed by a pair of a lower front U-shaped groove DLF which opens upward in the Z1 direction (predetermined direction) and an upper front U-shaped groove ULF which opens downward in the Z2 direction (direction opposite to the predetermined direction). More specifically, the lower front U-shaped groove DLF and the upper front U-shaped groove ULF are arranged to be adjacent to each other in the front to rear direction (the X-axis direction) and to oppose each other in the vertical direction. In a similar manner, the first rear through hole THB1 is formed by a pair of U-shaped grooves, that is, a lower rear U-shaped groove DLB which opens

upward in the Z1 direction (predetermined direction) and an upper rear U-shaped groove ULB which opens downward in the Z2 direction (direction opposite to the predetermined direction). More specifically, the lower rear U-shaped groove DLB and the upper rear U-shaped groove ULB are arranged to be adjacent to each other in the front to rear direction (the X-axis direction) and to oppose each other in the vertical direction.

(113) In addition, the first right-bearing portion **32AR** includes a first semi-circle cutout **CT1** through which the axis **5BX** of the second shaft member **5B** passes. In FIG. **11A**, the first semi-circle cutout **CT1** is formed by a combination of an upper front L-shaped groove **URF**, an upper rear L-shaped groove **URB**, and a lower L-shaped groove **DR**. The upper front L-shaped groove **URF** and the upper rear L-shaped groove **URB** open downward (in the Z2 direction) and rightward (in the Y2 direction). The lower L-shaped groove **DR** opens upward (in the Z1 direction) and rightward (in the Y2 direction). More specifically, the upper front L-shaped groove **URF** and the lower L-shaped groove **DR** are aligned in the front to rear direction (the X-axis direction). In a similar manner, the lower L-shaped groove **DR** and the upper rear L-shaped groove **URB** are aligned in the front to rear direction (the X-axis direction).

(114) In this manner, each of the first front through hole **THF1** and the first rear through hole **THB1** is formed by a pair of U-shaped grooves, and the first semi-circle cutout **CT1** is formed by a combination of L-shaped grooves. Hence, the first lens holder **3A** can be manufactured by injection molding using a pair of molds (an upper mold and a lower mold) that can be separated in the vertical direction and do not include a slide core. Therefore, such a configuration can reduce the manufacturing cost of the lens holder driving device **101**.

(115) As described above, the lens holder driving device **101** according to the embodiment of the present disclosure includes, as illustrated in FIGS. **3** and **4**, the fixed member **FB** (the base member **2**), the first lens holder **3A** which can hold the first lens body **LS1**, the second lens holder **3B** which can hold the second lens body **LS2** arranged to have the same optical axis **OA** as the first lens body **LS1**, the first shaft member **5A** and the second shaft member **5B** which are attached to the fixed member **FB** (the base member **2**) and can guide the movement of the first lens holder **3A** and the second lens holder **3B** in the optical-axis direction with respect to the fixed member **FB** (the base member **2**), the first piezoelectric driver **PD1** which includes the first piezoelectric element **8A** and moves the first lens holder **3A** along the optical-axis direction by the movement (circular motion) of the first piezoelectric element **8A**, and the second piezoelectric driver **PD2** which includes the second piezoelectric element **8B** and moves the second lens holder **3B** along the optical-axis direction by the movement (circular motion) of the second piezoelectric element **8B**. The first lens holder **3A** includes a portion (the first left-bearing portion **32AL**) near the first shaft member **5A** and a portion (the first right-bearing portion **32AR**) near the second shaft member **5B**. The first piezoelectric driver **PD1** is arranged at the portion (the first left-bearing portion **32AL**) near the first shaft member **5A** in the first lens holder **3A**, and contacts the first shaft member **5A**. The second lens holder **3B** includes a portion (the second left-bearing portion **32BL**) near the first shaft member **5A** and a portion (the second right-bearing portion **32BR**) near the second shaft member **5B**. The second piezoelectric driver **PD2** is arranged at the portion (the second right-bearing portion **32BR**) near the second shaft member **5B** in the second lens holder **3B**, and contacts the second shaft member **5B**.

(116) In this configuration, the movement of the first piezoelectric driver **PD1** and the movement of the second piezoelectric driver **PD2** can be prevented from interfering with each other via the same one axial member **5** when the first lens holder **3A** and the second lens holder **3B** are moved simultaneously in the optical-axis direction. This is because the first piezoelectric driver **PD1** is configured to contact the first shaft member **5A** and the second piezoelectric driver **PD2** is configured to contact the second shaft member **5B**. Hence, in this configuration, the first lens holder **3A** and the second lens holder **3B** can be moved simultaneously in a stable manner. In this case, the first lens holder **3A** and the second lens holder **3B** may move in the same direction or

move in opposite directions from each other.

(117) In addition, in the above-described embodiment, the pair of lens holders **3** (the first lens holder **3A** and the second lens holder **3B**) are formed as identical components that have the same shape and size. The respective pairs of the moving magnets **4**, the shaft members **5**, the magnetic sensors **6**, the piezoelectric elements **8**, the contact members **9**, the circuit boards **10**, the fixed magnets **11**, the magnetic members **12**, the biasing members **13**, the metal parts **14**, and the synthetic resin parts **15** are also formed to be identical components in a manner similar to the pair of lens holders **3**.

(118) This configuration can reduce the number of components used in the lens holder driving device **101**.

(119) In addition, as illustrated in FIG. **4** in the above-described embodiment, the first piezoelectric element **8A** may extend in a direction intersecting a direction (the X-axis direction) of the axis (the axis **5AX**) of the first shaft member **5A**. The second piezoelectric element **8B** may extend in a direction intersecting the direction of the axis (axis **5BX**) of the second shaft member **5B**.

(120) In this configuration, each shaft member **5** can be reliably brought into contact with the corresponding piezoelectric driver PD even when the shaft member **5** has a cylindrical shape. More specifically, in this configuration, the first shaft member **5A** can be reliably brought into contact with the first piezoelectric driver PD**1**, and the second shaft member **5B** can be reliably brought into contact with the second piezoelectric driver PD**2**. Note that in FIG. **4**, each piezoelectric element **8** is arranged to extend in the direction (the Z-axis direction) perpendicular to the direction (the X-axis direction) of the axis of the corresponding shaft member **5**. However, each piezoelectric element **8** may be arranged to extend in a direction intersecting the direction (the X-axis direction) of the axis of the corresponding shaft member **5** at an angle other than 90°. Reducing the length (height) of each piezoelectric element **8** in the Z-axis direction can reduce the length (height) of the lens holder driving device **101** in the Z-axis direction.

(121) Further, in the above-described embodiment, the first biasing member **13A** for biasing the first piezoelectric driver PD**1** toward the first shaft member **5A** may be provided in the first lens holder **3A** as illustrated in FIG. **4**. The second biasing member **13B** for biasing the second piezoelectric driver PD**2** toward the second shaft member **5B** may be provided in the second lens holder **3B**.

(122) This configuration allows the movement (circular motion) of each piezoelectric element **8** to be reliably transmitted to the corresponding shaft member **5**. More specifically, in this configuration, the movement of the first piezoelectric element **8A** can be reliably transmitted to the first shaft member **5A** and the movement of the second piezoelectric element **8B** can be reliably transmitted to the second shaft member **5B** while preventing interference between the respective movements of the first piezoelectric element **8A** and the second piezoelectric element **8B**.

(123) Further, as illustrated in FIGS. **6A** and **6B** in the above-described embodiment, the first biasing member **13A** may be made of a plate-shaped metal member such as a flat spring and include first fixing portions (the front fixing portion **13FF** and the rear fixing portion **13FB**) which are fixed to the first lens holder **3A**, two first support portions (the upper support portion **13SU** and the lower support portion **13SD**) for supporting two locations (the first node ND**1** and the second node ND**2**) which are spaced apart from each other on one side (the Y1 side) of the first piezoelectric driver PD**1**, and two first elastic deforming portions (the upper elastic deforming portion **13EU** and the lower elastic deforming portion **13ED**) which are capable of being elastically deformed and are provided between the first fixing portions (the front fixing portion **13FF** and rear fixing portion **13FB**) and the two first support portions (the upper support portion **13SU** and the lower support portion **13SD**). In this case, as illustrated in FIG. **6A**, the first shaft member **5A** may be in contact with the other side (the Y2 side) of the first piezoelectric driver PD**1** (the first contact member **9A**) in a space between the two first support portions (the upper support portion **13SU** and the lower support portion **13SD**). In a similar manner, as illustrated in FIG. **5**, the second biasing

member **13B** may be made of a plate-shaped metal member such as a flat spring, and may include second fixing portions (the front fixing portion **13FF** and the rear fixing portion **13FB**) which are fixed to the second lens holder **3B**, two second support portions (the upper support portion **13SU** and the lower support portion **13SD**) for supporting two locations spaced apart from each other on one side (the Y2 side) of the second piezoelectric driver **PD2**, and two second elastic deforming portions (the upper elastic deforming portion **13EU** and the lower elastic deforming portion **13ED**) which are capable of being elastically deformed and are provided between the second fixing portions (the front fixing portion **13FF** and the rear fixing portion **13FB**) and the two second support portions (the upper support portion **13SU** and the lower support portion **13SD**). In this case, the second shaft member **5B** may be in contact with the other side (the Y1 side) of the second piezoelectric driver **PD2** (the second contact member **9B**) in a space between the two second support portions (the upper support portion **13SU** and the lower support portion **13SD**).

(124) This configuration can prevent each biasing member **13** from obstructing the movement (circular motion) of the corresponding piezoelectric element **8**. This is because the areas near the two nodes of each piezoelectric element **8**, which vibrates (moves in a circular motion), are supported by the corresponding biasing member **13**.

(125) In addition, in the above-described embodiment, as illustrated in FIG. 4, the first piezoelectric driver **PD1** may be provided on the outside (the Y1 side) of the first shaft member **5A** away from the second shaft member **5B**, and the second piezoelectric driver **PD2** may be provided on the outside (the Y2 side) of the second shaft member **5B** away from the first shaft member **5A**.

(126) This configuration can improve the manufacturing productivity of the lens holder driving device **101**. This is because the piezoelectric driver **PD** can be attached easily to the lens holder **3** compared to a case in which the piezoelectric driver **PD** provided on the inner side of the shaft member **5**.

(127) In addition, in the above-described embodiment, as illustrated in FIG. 4, the first magnet (the first moving magnet **4A**) may be provided in the first lens holder **3A**. In this case, the first magnetic member that exerts a magnetic attraction force in the direction (the Z-axis direction) intersecting the optical-axis direction may be provided between the fixed member **FB** (the base member **2**) and the first magnet (the first moving magnet **4A**) over the entire range of movement of the first lens holder **3A**. Further, as illustrated in FIG. 4, the second magnet (the second moving magnet **4B**) may be provided in the second lens holder **3B**. In this case, the second magnetic member that exerts a magnetic attraction force in the direction (the Z-axis direction) intersecting the optical-axis direction may be provided between the fixed member **FB** (the base member **2**) and the second magnet (the second moving magnet **4B**) over the entire range of movement of the second lens holder **3B**. In FIG. 4, the first magnetic member is the second shaft member **5B** and the second magnetic member is the first shaft member **5A**. However, the first magnetic member may be an unillustrated member other than the second shaft member **5B** and the second magnetic member may be an unillustrated member other than the first shaft member **5A**.

(128) This configuration can restrict the rattling of the lens holders **3**. In the example illustrated in FIG. 8B, the magnetic attraction force that acts between the first moving magnet **4A** and the first magnetic member results in a torque (a biasing torque **TQ** indicated by a dash-dotted arrow) which causes the first lens holder **3A** to rotate about the axis (the axis **5AX**) of the first shaft member **5A**. The biasing torque **TQ** acts to press the right bearing portion **32AR** of the first lens holder **3A** against the second shaft member **5B** from the upper side (the Z1 side). In addition to the biasing torque **TQ**, a torque (dead-weight torque) from the dead weight of the first lens holder **3A** acts on the first lens holder **3A** to cause the first lens holder **3A** to rotate about the axis (axis **5AX**) of the first shaft member **5A**. The first moving magnet **4A** is configured such that the biasing torque **TQ** generated from the attraction force becomes higher than the dead-weight torque. Hence, whatever the posture of the lens holder driving device **101** may be (for example, even when the lens holder driving device **101** is turned upside down), the combined torque obtained by combining the biasing

torque TQ and the dead-weight torque can constantly act to press the first right-bearing portion **32AR** of the first lens holder **3A** against the second shaft member **5B** from the Z1 side. That is, the combined torque acts to press the first right-bearing portion **32AR** of the first lens holder **3A** against the second shaft member **5B** from the Z1 side not only when the biasing torque TQ and the dead-weight torque are in the same direction, but also when the biasing torque TQ and the dead-weight torque are in opposite directions to each other. As a result, the first right-bearing portion **32AR** of the first lens holder **3A** may be kept in constant contact with the second shaft member **5B** regardless of the posture of the lens holder driving device **101** (for example, even when the lens holder driving device **101** is turned upside down). Therefore, the rattling of the first right-bearing portion **32AR** and the second shaft member **5B** can be reduced. Rattling of the first left-bearing portion **32AL** of the first lens holder **3A** and the first shaft member **5A**, the rattling of second left-bearing portion **32BL** of the second lens holder **3B** and the first shaft member **5A**, and the rattling of the second right-bearing portion **32BR** of the second lens holder **3B** and the second shaft member **5B** can similarly be reduced. Furthermore, this configuration can improve the drop impact resistance of the lens holder driving device **101**.

(129) In addition, the first magnetic member that exerts an attraction force between itself and the first magnet (the first moving magnet **4A**) in the direction intersecting the optical-axis direction may be either the first shaft member **5A** or the second shaft member **5B**. In a similar manner, the second magnetic member that exerts an attraction force between itself and the second magnet (the second moving magnet **4B**) in the direction intersecting the optical-axis direction may be the other of the first shaft member **5A** or the second shaft member **5B**.

(130) In addition, in the above-described embodiment, as illustrated in FIG. 4, the first magnet (the first moving magnet **4A**) may be fixed to the portion (the first right-bearing portion **32AR**) near the second shaft member **5B** in the first lens holder **3A**. In this case, the first magnetic member may be the second shaft member **5B**. The second magnet (the second moving magnet **4B**) may be fixed to the portion (the second left-bearing portion **32BL**) near the first shaft member **5A** in the second lens holder **3B**. In this case, the second magnetic member may be the first shaft member **5A**.

(131) In this configuration, the magnetic member for generating an attraction force between itself and the moving magnet **4** need not be provided separately from the shaft member **5**. Hence, this configuration can reduce the manufacturing cost of the lens holder driving device **101**.

(132) Note that in a case in which the first shaft member **5A** is the first magnetic member, it is preferable to provide the first magnet (the first moving magnet **4A**) in a portion, of the first lens holder **3A**, which is near the first shaft member **5A**. In addition, in a case in which the second shaft member **5B** is the second magnetic member, it is preferable to provide the second magnet (the second moving magnet **4B**) in a portion, of the second lens holder **3B**, which is near the second shaft member **5B**.

(133) In addition, in the above-described embodiment, as illustrated in FIG. 4, the first piezoelectric driver PD1 may include the first contact member **9A** that is made of metal and contacts the first shaft member **5A**. Also, the second piezoelectric driver PD2 may include the second contact member **9B** that is made of metal and contacts the second shaft member **5B**. In this case, the first shaft member **5A** and the second shaft member **5B** may be made of a metal that is harder than that of the first contact member **9A** and the second contact member **9B**. That is, the first shaft member **5A** and the second shaft member **5B** may be made of a first metal, the first contact member **9A** and the second contact member **9B** may be made of a second metal, and the first metal may be harder than the second metal.

(134) This configuration helps reduce the wear of the shaft members **5** and the contact members **9**. In particular, this configuration reduces the wear of the shaft members **5** because the shaft members **5** are made of a metal which is harder than that of the contact members **9**. Furthermore, this configuration can reduce the debris that is generated due to the contact between the shaft members **5** and the contact members **9**, thus reducing the debris that reaches the image sensor IS.

(135) In addition, in the above-described embodiment, as illustrated in FIGS. 11A and 11B, the portion (the first left-bearing portion 32AL) of the first lens holder 3A that is near the first shaft member 5A may include a first front wall portion FW1, in which the first front through hole THF1 where the first shaft member 5A is inserted is formed, and a first rear wall portion BW1, in which the first rear through hole THB1 where the first shaft member 5A is inserted is formed. In this case, the first piezoelectric driver PD1 may be arranged inside a first space SP1 between the first front wall portion FW1 and the first rear wall portion BW1. The first space SP1 may be covered by the first cover CV1 (see FIG. 4) which is attached to the portion (the first left-bearing portion 32AL) of the first lens holder 3A which is near the first shaft member 5A.

(136) In a similar manner, as illustrated in FIG. 4, the portion (the second right-bearing portion 32BR) of the second lens holder 3B which is near the second shaft member 5B may include the second front wall FW2, in which a second front through hole THF2 where the second shaft member 5B is inserted is formed, and the second rear wall BW2, in which a second rear through hole THB2 where the second shaft member 5B is inserted is formed. In this case, the second piezoelectric driver PD2 may be arranged in the second space SP2 between the second front wall portion FW2 and the second rear wall portion BW2. The second space SP2 may be covered by the second cover CV2 which is attached to the portion (the second right-bearing portion 32BR) of the second lens holder 3B which is near the second shaft member 5B.

(137) In this configuration, the debris generated due to the contact between each shaft member 5 and the corresponding contact member 9 can be confined in a sealed space. More specifically, in this configuration, the debris generated due to the contact between the first shaft member 5A and the first contact member 9A can be confined in the first space SP1 enclosed by the first left-bearing portion 32AL (the first front wall portion FW1 and the first rear wall portion BW1) and the first cover CV1. Furthermore, in this configuration, the debris generated due to the contact between the second shaft member 5B and the second contact member 9B can be confined in the second space SP2 enclosed by the second right-bearing portion 32BR (the second front wall FW2 and the second rear wall BW2) and the second cover CV2.

(138) In addition, in the above-described embodiment, as illustrated in FIG. 3, the first magnetic field generating member MG1 and the second magnetic field generating member MG2 may be provided in the fixed member FB (the base member 2). In this case, as illustrated in FIG. 8B, the first magnetic field generating member MG1 may be arranged to face the portion (the first left-bearing portion 32AL) of the first lens holder 3A which is near the first shaft member 5A. The first magnetic sensor 6A for detecting the magnetic field from the first magnetic field generating member MG1 may be arranged near the portion (the first left-bearing portion 32AL) of the first lens holder 3A which is near the first shaft member 5A. Also, the second magnetic field generating member MG2 may be arranged to face the portion (the second right-bearing portion 32BR) of the second lens holder 3B which is near the second shaft member 5B. The second magnetic sensor 6B for detecting the magnetic field from the second magnetic field generating member MG2 may be provided in the portion (the second right-bearing portion 32BR) of the second lens holder 3B which is near the second shaft member 5B.

(139) This configuration can suppress the magnetic interference between the first magnetic field generating member MG1 and the second magnetic field generating member MG2. The magnetic interference can be suppressed because the first magnetic field generating member MG1 and the second magnetic field generating member MG2 are provided at positions apart from each other. Hence, in this configuration, the movement of the first lens holder 3A and the movement of the second lens holder 3B can be detected more accurately.

(140) In addition, in the above-described embodiment, as illustrated in FIG. 8A, the first magnetic field generating member MG1 may partially overlap with the second magnetic field generating member MG2 in the optical-axis direction. In FIG. 8A, the width WD1 is the length (width) of the first magnetic field generating member MG1 in the optical-axis direction, and the width WD2 is the

length (width) of the second magnetic field generating member MG2 in the optical-axis direction. The first magnetic field generating member MG1 overlaps with the second magnetic field generating member MG2 over the distance DS in the optical-axis direction.

(141) This configuration allows the sum of the length of the range of movement of the first lens holder 3A and the length of the range of movement of the second lens holder 3B to be set longer than the length of the lens holder driving device 101 in the optical-axis direction. Hence, in this configuration, it is possible to reduce the size and weight of the lens holder driving device 101 while allowing a desired range of movement to be implemented for each of the first movable member MB1 and the second movable member MB2.

(142) In addition, in the above-described embodiment, the first lens holder 3A may include the plurality of through holes (the first front through hole RHF1, the first rear through hole RHB1, the first front through hole THF1, and the first rear through hole THB1) extending in the optical-axis direction as illustrated in FIGS. 11A and 11B. The first front through hole RHF1 may be formed by the combination of the first groove (the lower front U-shaped groove DCF) which opens upward and the second groove (the upper front U-shaped groove UCF) which opens downward. The first rear through hole RHB1 may be formed by the combination of the first groove (the lower rear U-shaped groove DCB) which opens upward and the second groove (the upper rear U-shaped groove UCB) which opens downward. The first front through hole THF1 may be formed by the combination of the first groove (the lower front U-shaped groove DLF) which opens upward and the second groove (the upper front U-shaped groove ULF) which opens downward. The first rear through hole THB1 may be formed by the combination of the first groove (the lower rear U-shaped groove DLB) which opens upward and the second groove (the upper rear U-shaped groove ULB) which opens downward. The plurality of through holes (a second front through hole RHF2, a second rear through hole RHB2, the second front through hole THF2, and the second rear through hole THB2) formed in the second lens holder 3B illustrated in FIG. 4 may be formed in a similar manner to the above-described plurality of through holes formed in the first lens holder 3A. The first semi-circle cutout CT1 formed in the first lens holder 3A illustrated in FIGS. 11A and 11B and the second semi-circle cutout CT2 formed in the second lens holder 3B illustrated in FIG. 4 can also be formed in a manner similar to the above-described plurality of through holes formed in the first lens holder 3A.

(143) This configuration can reduce the manufacturing cost of the lens holder driving device 101. Each of the first front through hole THF1 and the first rear through hole THB1 is formed by a pair of U-shaped grooves, and the first semi-circle cutout CT1 is formed by a combination of L-shaped grooves. Hence, the first lens holder 3A can be manufactured by injection molding using a pair of dies (an upper die and a lower die) which can be separated in the vertical direction and do not include a slide core. In a similar manner, each of the second front through hole THF2 and the second rear through hole THB2 is formed by a pair of U-shaped grooves, and the second semi-circle cutout CT2 is formed by a combination of L-shaped grooves. Hence, the second lens holder 3B can be manufactured by injection molding using a pair of dies (an upper die and a lower die) which can be separated in the vertical direction and do not include a slide core. Note that since the first lens holder 3A and the second lens holder 3B are the same shape and size as illustrated in FIGS. 4, 11A, and 11B, they can be formed by the same set of molds.

(144) An embodiment of the present disclosure has been described above. However, the present disclosure is not limited to the embodiment described above. Various modifications and substitutions can be applied without departing from the spirit and scope of the present disclosure. The features described with reference to the above-described embodiment may be appropriately combined unless they are technically inconsistent.

(145) For example, in the above-described embodiment, the first through hole TH1 is formed by the combination of two U-shaped grooves as illustrated in FIGS. 11A and 11B. However, at least one of the U-shaped grooves may be a V-shaped groove. The first through hole RH1 (see FIGS.

11A and 11B), the second through hole TH2 (see FIG. 4), and a second through hole RH2 (see FIG. 4) may also be formed such that at least one of the U-shaped grooves may be a V-shaped groove. (146) In addition, in the above-described embodiment, the first moving magnet 4A is fixed to the portion near the second shaft member 5B in the first lens holder 3A. However, the first moving magnet 4A may be fixed to both the portion near the first shaft member 5A and the portion near the second shaft member 5B. The second moving magnet 4B may be formed in a similar manner. Furthermore, the first moving magnet 4A may be fixed to the portion near the second shaft member 5B in the first lens holder 3A, and the second moving magnet 4B may be fixed to the portion near the second shaft member 5B in the second lens holder 3B.

Claims

1. A lens holder driving device comprising: a fixed member; a first lens holder configured to hold a first lens body; a second lens holder configured to hold a second lens body arranged to have a same optical axis as the first lens body; a first shaft member and a second shaft member provided in the fixed member and configured to guide the first lens holder and the second lens holder to move in an optical-axis direction with respect to the fixed member; a first piezoelectric driver including a first piezoelectric element and configured to move the first lens holder along the optical-axis direction based on movement of the first piezoelectric element; and a second piezoelectric driver including a second piezoelectric element and configured to move the second lens holder along the optical-axis direction based on movement of the second piezoelectric element, wherein the first lens holder includes a first portion that is positioned closer to the first shaft member than to the second shaft member and a second portion that is positioned closer to the second shaft member than to the first shaft member, the first piezoelectric driver is provided in the first portion in the first lens holder such that the first piezoelectric driver directly contacts the first shaft member, while the first piezoelectric driver is arranged so as not to contact the second shaft member, the second lens holder includes a third portion that is positioned closer to the first shaft member than to the second shaft member and a fourth portion that is positioned closer to the second shaft member than to the first shaft member, and the second piezoelectric driver is provided in the fourth portion in the second lens holder such that the second piezoelectric driver directly contacts the second shaft member, while the second piezoelectric driver is arranged so as not to contact the first shaft member, wherein the first piezoelectric element extends in a direction intersecting a direction of an axis of the first shaft member, and the second piezoelectric element extends in a direction intersecting a direction of an axis of the second shaft member, wherein the first lens holder includes a first biasing member configured to bias the first piezoelectric driver toward the first shaft member, and the second lens holder includes a second biasing member configured to bias the second piezoelectric driver toward the second shaft member, wherein the first biasing member is a plate-shaped metal member and includes: a first fixing portion fixed to the first lens holder; two first support portions configured to support two locations which are spaced apart from each other on one side of the first piezoelectric driver, said two first support portions being arranged to be parallel to each other; and two first elastic deforming portions capable of being elastically deformed and provided between the first fixing portion and the two first support portions, wherein the other side of the first piezoelectric driver contacts the first shaft member in a space between the two first support portions, the second biasing member is a plate-shaped metal member and includes: a second fixing portion fixed to the second lens holder; two second support portions configured to support two locations which are spaced apart from each other on one side of the second piezoelectric driver, said two second support portions being arranged to be parallel to each other; and two second elastic deforming portions provided between the second fixing portion and the two second support portions and capable of being elastically deformed, wherein the other side of the second piezoelectric driver contacts the second shaft member in a space between the two second

support portions, the first fixing portion, the two first elastic deforming portions, and the two first support portions integrally form the first biasing member as a single piece, and the second fixing portion, the two second elastic deforming portions, and the two second support portions integrally form the second biasing member as a single piece.

2. The lens holder driving device according to claim 1, wherein the first shaft member and the second shaft member are spaced apart from each other, the first shaft member is positioned between the first piezoelectric driver and the second shaft member, and the second shaft member is positioned between the second piezoelectric driver and the first shaft member.

3. The lens holder driving device according to claim 1, wherein the first lens holder includes a first magnet, the fixed member includes a first magnetic member configured to exert an attraction force between the fixed member and the first magnet in a direction intersecting the optical-axis direction over an entire range of the movement of the first lens holder, the second lens holder includes a second magnet, and the fixed member includes a second magnetic member configured to exert an attraction force between the fixed member and the second magnet in a direction intersecting the optical-axis direction over an entire range of movement of the second lens holder.

4. The lens holder driving device according to claim 3, wherein the first magnetic member includes either the first shaft member or the second shaft member, and the second magnetic member includes the other of the first shaft member or the second shaft member.

5. The lens holder driving device according to claim 3, wherein the first magnet is fixed to the second portion in the first lens holder, the first magnetic member is the second shaft member, the second magnet is fixed to the third portion in the second lens holder, and the second magnetic member is the first shaft member.

6. The lens holder driving device according to claim 1, wherein the first piezoelectric driver includes a first contact member, which is made of a metal and is configured to contact the first shaft member, the second piezoelectric driver includes a second contact member, which is made of a metal and is configured to contact the second shaft member, and the first shaft member and the second shaft member are made of a first metal and the first contact member and the second contact member are made of a second metal, wherein the first metal is harder than the second metal.

7. The lens holder driving device according to claim 1, wherein the first portion in the first lens holder includes: a first front wall portion, which includes a first front through-hole where the first shaft member is inserted; and a first rear wall portion, which includes a first rear through-hole where the first shaft member is inserted, the first piezoelectric driver is arranged in a first space between the first front wall portion and the first rear wall portion, the first space is covered by a first cover attached to the first portion in the first lens holder, and the fourth portion in the second lens holder includes: a second front wall portion, which includes a second front through-hole where the second shaft member is inserted; and a second rear wall portion, which includes a second rear through-hole where the second shaft member is inserted, the second piezoelectric driver is arranged in a second space between the second front wall portion and the second rear wall portion, and the second space is covered by a second cover attached to the fourth portion in the second lens holder.

8. The lens holder driving device according to claim 1, wherein the fixed member includes a first magnetic field generating member and a second magnetic field generating member, the first magnetic field generating member faces the first portion in the first lens holder, the first portion in the first lens holder includes a first magnetic sensor configured to detect a magnetic field from the first magnetic field generating member, the second magnetic field generating member faces the fourth portion in the second lens holder, and the fourth portion in the second lens holder includes a second magnetic sensor configured to detect a magnetic field from the second magnetic field generating member.

9. The lens holder driving device according to claim 8, wherein the first magnetic field generating member partially overlaps with the second magnetic field generating member in the optical-axis direction.

10. The lens holder driving device according to claim 1, wherein each of the first lens holder and the second lens holder includes a plurality of through holes that run in the optical-axis direction, and each of the plurality of through holes is formed by a combination of a first groove which opens in a predetermined direction and a second groove which opens in a direction opposite to the predetermined direction.

11. The lens holder driving device according to claim 1, wherein each of the two first support portions includes a first recessed portion that is configured to hold the first piezoelectric driver, and each of the two second support portions includes a second recessed portion that is configured to hold the second piezoelectric driver.
